

Implementation of Right-in threaded Binary tree :
The implementation of a right-in threaded binary tree under dynamic node implementation, an extra logical field rthread is included within each node to indicate whether or not its right pointer is a thread. ie, the rthread is TRUE if right pointer is NULL or a non NULL thread.

Note: The rthread field of the right most node of a tree (the last node in the inorder traversal) is also set to TRUE, although its right field remains NULL.

```
Struct node
{
int info;
Struct node *right, *left;
int rthread;
};
typedef Struct node *Nodepiter;
```

The function to implment inorder traversal of a right-in threaded binary tree:

```
void inorder(Nodepiter root)
{
Nodepiter p, q;
p=root;
do
{
q=null;
while(p!=NULL)
{
q=p;
p=p->left;
}
if(q!=NULL)
{
printf("%d\t", q->info);
p=q->right;
}
```

```

        while(q->rthread && p!=NULL)
        {
            printf("%d\t", p->info);
            q=p;
            p=p->right;
        }
    } while(q!=NULL);
}

```

Creation of Right-in threaded binary tree:

```

Nodeptr maketree(int x)
{
    Nodeptr p;
    p=getnode();
    p->info=x;
    p->left=NULL;
    p->right=NULL;
    p->rthread=TRUE;
    return(p);
}

```

```

void setleft(Nodeptr p, int x)
{
    Nodeptr q;
    if(p==NULL) printf(" Invalid insertion");
    else if (p->left !=NULL)
        printf("Invalid Insertion");
    else
    {
        q= getnode();
        q->info=x;
        q->left=NULL;
        p->left=q;
        q->right=p; /*inorder successor of q is p*/
        q->rthread=TRUE;
    }
}

```

```
    }  
}
```

```
void setright(Nodeptr p, int x)  
{  
    Nodeptr q, temp;  
    if(p==NULL) printf(" Invalid insertion");  
    else if (!p->rthread)  
        printf("Invalid Insertion");  
    else  
    {  
        q= getnode();  
        q->info=x;  
        q->left=NULL;  
        temp=p->right; /*Save the inorder successor  
of node p*/  
        p->right=q;  
        p->rthread =FALSE;  
        q->right=temp;  
        q->rthread=TRUE;  
    }  
}
```